



CARTRIDGE BOX FOR REPEATING RIFLES (1871)
Manufacturers of firearms preferred owners to use their own brand of ammunition. This pack of Winchester rifle cartridges is typical of the late 1800s.



BULLET BOX FOR MATCH RIFLES (1872)
To maintain consistent shooting, competitors in long-range “match” rifle-shooting contests demanded great precision in the manufacture of ammunition components. Swaged, or pressure-formed, bullets were individually weighed.

TOOLS



BALL MOLD



BALL/BULLET MOLD

CASTING BULLETS

Until the sale of loose bullets became common, firearms were supplied with molds, with cavities into which molten lead was poured via channels. The metal solidified in the molds, producing ammunition of the correct size. Excess metal that solidified in the channels was termed sprue. Seen here are two molds. The upper mold has an automatic sprue cutter, which simply sliced off the sprue as the mold was opened. The lower mold has a more usual pivoting sprue cutter, which would cut off the sprue when knocked to one side.



**CAPPER/
DECAPPER**

RELOADING PRIMERS

This tool was used to remove fired primers—a special form of percussion cap—and install fresh ones in the heads of metallic center-fire cartridges.

Pistol cartridges

Pistols fire over a shorter range than rifles, and they use shorter cartridges that contain less powder and are less powerful. Shorter barrels mean a lower bullet velocity and lower penetrating power. Like rifle cartridges, they developed from rim-fire to the better center-fire design in the 1860s.



.44IN HENRY (1860)
This rim-fire round had primer arranged around the base of its case. It was soon superseded by the center-fire cartridge.



.44IN ALLEN AND WHEELOCK (1860s)
Allen and Wheelock revolvers were chambered for “lip-fire” cartridges (similar to rim-fire), chiefly in small calibers.



.45IN COLT (BÉNÉT 1865)
Colonel SV Bénét’s 1865 version of the center-fire cartridge formed the basis for Berdan’s popular center-fire metallic cartridge.



.45IN COLT (THUER 1868)
Alexander Thuer developed a method of converting Colt “cap-and-ball” revolvers to fire this tapering brass cartridge.



.44IN SMITH AND WESSON RUSSIAN (1870)
This center-fire cartridge was supplied to the Russian Army for Smith and Wesson revolvers.



.577IN WEBLEY (1880s)
Many small-caliber cartridges lacked the explosive power to stop a man. Webley addressed this with a .577in cartridge.



.476IN WEBLEY (1881)
The .577in revolver was unwieldy and a replacement in .476in caliber was adopted instead. It, too, was short-lived.



10.4MM BODEO (1889)
This revolver cartridge, used by the Italian Army from 1891, produced a muzzle velocity of 837ft (255m) per second—higher than most cartridges of the time.



.455IN WEBLEY (1891)
Webley’s first smokeless-powder cartridge was more powerful than earlier types. As a lighter bullet with a more powerful charge, it could travel faster and inflict more damage.



7.63MM BERGMANN (1896)
The first cartridges made for the Bergmann No 3 pistol in 1896 were rimless and grooveless, with a sharp nose.

Shotgun cartridges

Only the very largest shotgun cartridges were made entirely of brass. Others had cardboard bodies.



10-BORE PIN-FIRE
Production of most pin-fire cartridges ceased in the 1860s. However those for shotguns, continued until the 1920s.

WILDFOWL CARTRIDGE

Large cartridges such as this were loaded with up to ¾oz (20g) of gunpowder and 3½oz (100g) of shot.